

llmXive Automated Review of Code2LoRA: Hypernetwork-Generated Adapters for Code Language Models under Software Evolution

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a novel hypernetwork approach for generating LoRA adapters, and the core claims regarding the architecture and the benchmark construction appear internally consistent. However, there are specific instances where the strength of the claims exceeds the provided evidence or requires clarification to avoid misinterpretation.

First, the claim in Section 5.2 and Table 3 that Code2LoRA Evo (64.5% IR EM) exceeds the per-repo LoRA (pLoRA) upper bound (64.2% IR EM) is scientifically problematic. By definition, a model trained on the exact target repository’s data (pLoRA) should represent the performance ceiling for that specific data distribution. Surpassing this bound implies either a methodological flaw (e.g., the pLoRA baseline was not trained on the full test set, or there is data leakage in the Code2LoRA training), or a misunderstanding of the “upper bound” definition. If pLoRA was trained on a subset of the repository data while Code2LoRA utilized cross-repo transfer, the comparison is not a true “upper bound” test. This requires a rigorous explanation or correction of the baseline setup to ensure the claim of “exceeding the upper bound” is not misleading.

Second, the Out-of-Distribution (OOD) results in Section 5.3 and Table 4 rely on Exact Match (EM) as the primary metric. The authors correctly identify in the Limitations and Appendix that OOD targets are significantly shorter (median 7 chars vs 12-13), which artificially inflates EM scores. While the paper notes that Code2LoRA Evo leads by +1.8pp, the absolute EM values (74.1% vs 72.3%) are heavily influenced by this artifact. The claim that the model “remains best” is supported, but the presentation of the absolute performance gap as a definitive measure of superiority without length-normalized metrics (e.g., EditSim or CodeBLEU as primary OOD metrics) overstates the evidence. The text should explicitly state that the 1.8pp lead is a lower bound on the true performance gap, or provide a length-controlled analysis.

Finally, in Appendix A.4, the claim that “hallucinations/empty <1%” is too vague for a quantitative result. With 2,321 total errors, the authors should provide the exact count or percentage (e.g., “0.3%”) to substantiate the claim that this error mode is negligible.

These issues do not invalidate the core contribution but require clarification to ensure the accuracy of the reported performance metrics and the validity of the “upper bound” claim.

Action items.

- **[science]** Section 5.2 claims Code2LoRA Evo (64.5% IR EM) exceeds the per-repo LoRA (pLoRA) upper bound (64.2%). Since pLoRA trains on the target repo, exceeding it implies data leakage or a definition mismatch. Clarify how a generalization method beats a model trained on the exact target distribution.
- **[writing]** Section 5.3 notes OOD targets are shorter, inflating EM. The claim that Code2LoRA Evo leads by +1.8pp is valid, but the absolute EM gap (74.1% vs 72.3%) is confounded by length. Provide length-normalized metrics or clarify that the gap is a lower bound on true

capability.

- **[writing]** Appendix A.4 states ‘hallucinations/empty <1%’. With 2,321 errors, provide the exact count or percentage (e.g., 0.3%) to substantiate this quantitative claim. Vague bounds are insufficient for results sections.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure provides a clear visual overview of the architecture, but the caption contains a critical omission where the model name is missing before the possessive ‘s’, and the visual highlight in the input box is not explained.

Figure 2

The figure effectively visualizes the bursty nature of commits across five repositories, but the caption relies on an undefined variable (‘num-repos-total’) and the legend’s reference to ‘#QnAs’ is not explicitly linked to the ‘test-touching’ concept mentioned in the text.

Figure 3

The figure effectively visualizes the token length distributions, but the left panel’s legend includes a ‘4096 tokens’ entry that is not visible within the plot’s x-axis range, creating a minor inconsistency. Additionally, the caption’s description of the vertical lines is slightly less precise than the specific values provided in the legend.

Figure 4

The figure is a clear violin plot, but the caption is severely broken with missing method names and statistical symbols, rendering the specific performance claims and comparisons unreadable.

Figure 5

The figure contains a significant contradiction between the caption’s claim of underperformance for small datasets and the visual data showing high EM scores. Additionally, the x-axis label is malformed and the legend line style does not match the plot element.

Figure 6

Figure 6 presents scaling trends but the caption is incomplete (omits the error rate panel) and the ‘log-linear’ claim is not strongly supported by the data or fit statistics shown.

Figure 7

The figure effectively displays the trend of exact-match accuracy over normalized commit positions for multiple methods. However, the caption fails to define the specific method names (e.g., Code2LoRA-Evo) shown in the legend, and there is a minor inconsistency between the legend line styles and the plot lines.

Figure 8

The figure effectively visualizes the t-SNE clustering and color-coding described in the caption, but the text labels for repositories are too small and cluttered to be legible, and the colorbar label is poorly oriented.

Figure 9

Figure 9 effectively compares per-module weight norms between two methods, but lacks clarity on the meaning of percentages in the top heatmap’s y-axis labels and does not define FFT+DRC. Additionally, the colorbar scales between the two heatmaps may be inconsistent.

Figure 10

Figure 10 is a clear and well-constructed horizontal bar chart that effectively visualizes the error classification data. The axes are labeled with units, the bars include precise numerical values and percentages, and the caption accurately describes the content and taxonomy used.

Action items.

- **[science]** Figure 1: The caption references ‘s static hypernetwork (b) and ‘s recurrent hypernetwork (c), but the model name is missing from the text, making the figure labels ambiguous.
- **[writing]** Figure 1: The ‘Input’ box contains a code snippet with a yellow highlight on ‘??’, but the caption does not explain this visual cue or the specific task of assertion completion in detail.
- **[writing]** Figure 2: The caption references ‘num-repos-total’ as a variable rather than providing the actual count of repositories in the dataset.
- **[science]** Figure 2: The legend defines red dot area as proportional to ‘#QnAs’, but the caption claims the figure illustrates ‘Test-touching commits’; the figure fails to explicitly label the red dots as ‘test-touching’ or clarify if QnAs are the proxy for these commits.
- **[writing]** Figure 2: The x-axis label ‘Commit date (first-parent)’ is ambiguous; it is unclear if this represents the date of the first parent commit in a merge or simply the chronological

date of the commit itself.

- **[science]** Figure 3: The left panel legend lists ‘4096 tokens’ (orange dashed line), but no corresponding orange line is visible in the plot area, as the x-axis limit (4000) cuts it off. This creates a discrepancy between the legend and the visual data.
- **[writing]** Figure 3: The caption states that vertical dashed lines mark ‘common context window sizes,’ but the legend explicitly labels these lines with specific token counts (e.g., ‘512 tokens’, ‘2048 tokens’). The legend should be updated to reflect that these lines represent specific context window thresholds rather than just generic sizes.
- **[fatal]** Figure 4: The caption contains broken text where method names are missing (e.g., ‘of (Table checkpoints’, ‘and (median 66.7%)’, ‘for and’). The figure labels ‘Code2LoRA-Static’ and ‘Code2LoRA-Evo’ correspond to these missing names, but the caption fails to explicitly link them, making the specific performance claims unreadable.
- **[science]** Figure 4: The caption claims to report standard deviation (e.g., ‘=16.8’) for the methods, but the symbol preceding the value is missing/blank in the text. Without the symbol (e.g., σ or *SD*), the statistical metric is undefined.
- **[science]** Figure 5: The caption claims repositories with fewer than 50 training pairs ‘frequently underperform the IR-test pretrained baseline (46.8%)’, but the plot shows a dense cluster of points at 100% EM for $N < 50$, directly contradicting the text.
- **[writing]** Figure 5: The x-axis label ‘ $3 \times 10^{14} \times 10^1$ ’ is malformed and unreadable; it likely intends to show a range (e.g., 3×10^1 to 4×10^1) but the formatting is broken.
- **[writing]** Figure 5: The legend entry ‘ $N_{\text{train}} = 50$ ’ uses a dotted line style that does not match the vertical dotted line shown on the plot, creating ambiguity about what the line represents.
- **[science]** Figure 6: The caption claims performance improves ‘log-linearly’, but the left plot shows a log-linear fit (red dashed) with $R^2=0.721$ and a power-law fit (green solid) with $R^2=0.719$. The fits are nearly identical, and the data points show significant scatter (e.g., at 10^2 and 10^3 repositories), making the ‘log-linear’ claim weak and potentially misleading without statistical justification.
- **[science]** Figure 6: The right plot (‘Error Rate Scaling’) is not mentioned in the caption. The caption only describes ‘CR-test EM as a function of training repository count’, but the figure includes a second panel showing error rate, which is a different metric and should be described.
- **[writing]** Figure 6: The caption uses ‘benefits from repository diversity’ without specifying the method name (e.g., Code2LoRA), making it unclear which approach is being discussed. The method name should be explicitly stated.
- **[science]** Figure 7: The legend lists ‘Code2LoRA’ (blue dashed line with triangles) and ‘Code2LoRA-Evo’ (green solid line with stars), but the caption only describes the plot as

‘CR-test exact-match vs. normalized commit position’ without defining these specific method names or distinguishing between the ‘Evo’ and non-‘Evo’ variants.

- **[writing]** Figure 7: The legend uses inconsistent line styles to represent methods; ‘Text2LoRA’ is shown with a dash-dot line, ‘Code2LoRA’ with a dashed line, and ‘Code2LoRA-Evo’ with a solid line, but the legend markers (e.g., the dash-dot pattern for Text2LoRA) do not perfectly match the line styles in the plot area, potentially causing confusion.
- **[writing]** Figure 8: The colorbar label ‘CR Test Exact Match (%)’ is rotated 90 degrees and runs vertically along the axis, making it difficult to read; it should be horizontal or clearly legible.
- **[writing]** Figure 8: The text labels for individual repositories (e.g., ‘django’, ‘pymc’) are extremely small and densely packed, causing significant overlap and illegibility for many points.
- **[writing]** Figure 9: The top heatmap’s y-axis labels include percentages (e.g., ‘mkslides (12%)’) that are not defined in the caption or figure; clarify what these percentages represent.
- **[science]** Figure 9: The bottom heatmap’s colorbar scale (0.0–1.0) differs from the top heatmap’s implied scale (0.0–1.0 but visually compressed); ensure consistent normalization or explicitly state if scales differ.
- **[writing]** Figure 9: The caption refers to ‘FFT+DRC’ in the bottom panel but does not define what FFT+DRC is; provide a brief explanation or cross-reference to its definition elsewhere in the paper.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript relies heavily on domain-specific acronyms and abbreviations that are not defined at their first occurrence, creating unnecessary friction for non-specialist readers.

In Section 1 (Introduction), the terms “CR” (Cross-Repo) and “IR” (In-Repo) are introduced in a list but immediately used in the results summary without a clear declaration that they will serve as standard abbreviations. Similarly, “OOD” (Out-of-Distribution) appears in the introduction and throughout the results (Section 5.3) without being spelled out first.

Section 3.5 (Training) and Appendix B.1 introduce “BPTT” (Backpropagation Through Time) without expansion. While standard in deep learning, the paper’s scope includes software evolution, a field where this specific training acronym may not be universally known.

The term “DRC” (Dependency-Resolved Context) is used extensively in Section 6 and Appendix B.1 as a proper noun for a specific retrieval method, yet the acronym is never defined. This forces the reader to infer the meaning from context or search the bibliography, which is

poor practice for a general audience.

Finally, the abbreviation “pp” is used repeatedly in Section 1 and Section 7.1 to denote “percentage points.” While concise, this is technical jargon that should be written out in full to ensure clarity for all readers.

These issues are fixable by adding brief parenthetical definitions at the first instance of each term and ensuring a consistent “hereafter referred to as” convention for acronyms used throughout the text.

Action items.

- **[writing]** Define ‘BPTT’ (Backpropagation Through Time) at first use in Section 3.5 or Appendix B.1. Currently, the acronym appears without expansion, which excludes readers unfamiliar with recurrent training dynamics.”
- **[writing]** Replace ‘OOD’ with ‘out-of-distribution’ at first mention in Section 1 and Section 5.3. The acronym is used frequently without prior definition, creating a barrier for non-specialist readers.”
- **[writing]** Define ‘DRC’ (Dependency-Resolved Context) explicitly upon first introduction in Section 6 or Appendix B.1. The term is used as a proper noun without explanation, assuming reader familiarity with AST-based retrieval methods.”
- **[writing]** Clarify ‘CR’ and ‘IR’ (Cross-Repo, In-Repo) in Section 1 or Section 5.1. While defined in a list, the acronyms are used immediately in the results summary without a clear ‘hereafter referred to as’ statement.”
- **[writing]** Replace ‘pp’ (percentage points) with the full phrase ‘percentage points’ in Section 1 and Section 7.1. While common in statistics, it is jargon that should be spelled out for general accessibility.”

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a compelling architecture for generating LoRA adapters via hypernetworks, but several logical gaps undermine the strength of the conclusions drawn from the experimental data.

First, the claim in Section 3.1 and Table 1 that **Code2LoRA-Static** achieves an In-Repo (IR) Exact Match (EM) of 66.2%, effectively “matching” the **per-repo LoRA** (pLoRA) upper bound of 64.0% (or 64.2% in the evolution track table), is logically suspect. By definition, pLoRA is trained on the specific repository’s data, while Code2LoRA is trained on a cross-repo dataset (409 repos) and generates weights for a held-out repo. For a cross-repo model to equal the in-distribution upper bound without seeing the target repo’s training data implies either a flaw

in the experimental setup (e.g., data leakage where the target repo was in the training set) or a statistical anomaly that requires deeper statistical validation (e.g., significance testing) rather than a direct point-comparison. The text treats this as a definitive success without addressing the logical impossibility of a general model outperforming or equaling a specialized model on its own data without specific adaptation.

Second, the Out-of-Distribution (OOD) analysis in Section 5.3 and Appendix A.4 contains a circular argument regarding the validity of the EM metric. The authors explicitly state that OOD targets are significantly shorter (median 7 chars vs. 12-13 chars), which “inflates exact-match credit.” They then proceed to claim Code2LoRA is superior on OOD because it leads by +1.8pp in EM. If the metric is known to be biased by target length, comparing absolute EM scores between OOD and in-distribution sets is logically invalid. The conclusion that the model generalizes better is not supported by the provided evidence unless the authors normalize for target length or use a length-agnostic metric (like CodeBLEU or EditSim) as the primary claim for OOD superiority. The current logic relies on a metric they admit is flawed for the specific comparison.

Finally, the causal link between the “bursty commit pattern” (Figure 2) and the necessity of the GRU mechanism in **Code2LoRA-Evo** is asserted but not mechanistically explained. The paper argues that bursty commits cause staleness, necessitating a recurrent model. However, the GRU processes diffs sequentially. If commits are “bursty” (many occurring in a short time), the logical question is how the GRU handles the temporal density. Does it process them in arbitrary order? Does it aggregate them? The paper does not detail how the GRU’s hidden state updates specifically mitigate the “burstiness” compared to a simple averaging of diffs or a static snapshot of the latest commit. Without this mechanistic explanation, the claim that the GRU is the specific solution to bursty evolution remains an unsupported causal assertion.

Action items.

- **[writing]** The paper presents a compelling architecture for generating LoRA adapters via hypernetworks, but several logical gaps undermine the strength of the conclusions drawn from the experimental data. First, the claim in Section 3.1 and Table 1 that Code2LoRA-Static achieves an In-Repo (IR) Exact Match (EM) of 66.2%, effectively “matching” the per-repo LoRA (pLoRA) upper bound of 64.0% (or 64.2% in the evolution track table), is logically suspect. By definition, pLoRA is trained on the specific reposit

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extrapolate beyond the immediate evidence provided in the experiments, specifically regarding the nature of performance bounds and the universality of the proposed method’s superiority.

First, the assertion in Section 5.2 that `\codeloraevof` “exceeds the `\UseMacro{method-plora}` upper bound on IR” is a significant overreach. The term “upper bound” typically implies the theoretical maximum performance achievable given the data distribution. Since `\UseMacro{method-plora}` (per-repo LoRA) is trained directly on the specific repository’s training split, it represents the empirical ceiling for that specific data. Claiming a cross-repo trained model (`\codeloraevof`) surpasses this without per-repo adaptation suggests a logical inconsistency unless the “upper bound” is redefined (e.g., as a constraint on training time or parameter budget). Without clarifying that the comparison is against a *constrained* per-repo baseline rather than the *optimal* per-repo baseline, this claim misleads the reader about the model’s capabilities.

Second, the conclusion that “parametric adaptation outperforms context injection” (Section 5.2) is too broad. The context injection baselines (RAG, DRC) are evaluated with specific, fixed hyperparameters (e.g., $k=3$, 512 tokens). The paper does not explore whether optimizing retrieval parameters (e.g., larger k , dynamic context windows, or better chunking) could close the performance gap. By not testing the *best possible* context injection setup, the authors cannot definitively claim that parametric methods are inherently superior; they have only shown superiority over *suboptimal* context injection configurations.

Finally, the claims regarding Out-of-Distribution (OOD) generalization (Section 5.3) are overstated. The authors explicitly acknowledge in the Limitations and Appendix that OOD targets are significantly shorter (median 7 chars vs 12-13), which artificially inflates Exact Match scores. Despite this, the paper presents the OOD EM lead as evidence of the model’s robustness to temporal shifts. A claim of “generalization” should ideally be decoupled from such confounding artifacts. The paper should either normalize the metrics for target length or temper the conclusion to reflect that the observed OOD advantage may be partially driven by the easier target distribution rather than true semantic generalization.

Action items.

- **[science]** Claiming to exceed the ‘per-repo LoRA upper bound’ (Sec 5.2) is overreaching. Per-repo LoRA is trained on target data; claiming a cross-repo model beats it without qualification contradicts learning theory. Clarify if ‘upper bound’ refers to a specific constraint (e.g., time) or rephrase.
- **[science]** The claim that ‘parametric adaptation outperforms context injection’ (Sec 5.2) is too broad. Baselines use fixed retrieval params ($k=3$). Without testing optimal context injection, the paper cannot claim inherent superiority, only superiority over suboptimal baselines.
- **[science]** OOD generalization claims (Sec 5.3) are overstated. OOD targets are shorter (7 vs 12 chars), inflating EM. The paper admits this but still presents the EM lead as proof of robustness. Normalize metrics or temper the conclusion to address this confound.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses safety and ethics primarily through a brief “Potential risks” paragraph in the Limitations section and a “Privacy and Content Review” note in the Appendix. While the authors correctly identify that the dataset consists of public, permissively licensed repositories (MIT/Apache), the current treatment of safety concerns is insufficient for a paper proposing a model that deeply internalizes repository-specific logic.

First, regarding data privacy, the Appendix (Section `app:dataset:privacy`) explicitly states: “No PII scrubbing performed; dataset is redistribution of public source.” This is a significant oversight. Public GitHub repositories frequently contain accidentally committed secrets (API keys, tokens), personal email addresses in commit logs, or other sensitive information. By training a hypernetwork that generates adapters based on the entire repository context, the model risks memorizing and regurgitating this sensitive data. The authors must either confirm that a PII/secret scanning step was performed on the dataset prior to training or add a strong warning in the Limitations section advising users to audit the training data and the model’s outputs for sensitive information before deployment.

Second, the “Potential risks” discussion in the Limitations section (Section `sec:conclusion`) is too vague. It states that downstream risks “mirror standard code LLMs (insecure/incorrect completions).” However, the paper’s core contribution is the ability to adapt to specific repository conventions and internal logic (e.g., internal identifiers, specific error handling patterns). This capability introduces a unique risk of “contextual leakage,” where the model might inadvertently reproduce proprietary code patterns or sensitive business logic from the training set in a different context. The authors should expand this section to discuss the potential for intellectual property leakage and the specific risks of the model learning and reproducing insecure coding patterns present in the training repositories.

Finally, while the authors mention using LLMs for grammar polishing (Section `app:llms`), they do not disclose if any LLMs were used to generate the assertion-completion pairs or to filter the dataset. If LLMs were involved in dataset construction, this should be disclosed to ensure transparency regarding potential biases or hallucinations introduced during data creation.

Action items.

- **[writing]** The ‘Privacy and Content Review’ section (Appendix) explicitly states ‘No PII scrubbing performed.’ Given the dataset includes full repositories with commit history, there is a non-negligible risk of inadvertently including sensitive data (API keys, credentials, or personal emails) scraped from public GitHub. Authors must add a statement confirming that a PII scan was performed or provide a mechanism for users to filter such data.
- **[writing]** The ‘Potential risks’ section (Limitations) is overly generic, stating risks ‘mirror standard code LLMs.’ The paper claims the model internalizes repository-specific conventions

and identifiers. Authors should explicitly address the risk of the model memorizing and regurgitating proprietary code patterns or sensitive logic from the training repos, potentially facilitating IP leakage or security vulnerabilities in downstream applications.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a compelling hypernetwork architecture for generating repository-specific LoRA adapters, with a clear motivation regarding software evolution. The experimental design covers static and dynamic scenarios, and the dataset construction is substantial. However, the strength of the scientific evidence is currently undermined by three specific issues regarding experimental controls and statistical rigor.

First, the Out-of-Distribution (OOD) evaluation (Section 5.3, Table 3) suffers from a significant confound. The authors note in the Limitations that OOD targets are shorter (median 7 chars vs 12-13), which artificially inflates Exact Match (EM) scores. While they claim a +1.8pp lead for `\codeloraev0{}{}{}`, this margin is small and likely within the noise introduced by the length bias. The paper fails to provide a length-normalized analysis or a statistical significance test (e.g., bootstrapped confidence intervals) to prove the generalization advantage is real and not an artifact of the evaluation metric’s sensitivity to target length.

Second, the comparison between the Static and Evolution tracks lacks a controlled variable for data volume. The Evolution track training set contains 45,516 commits, whereas the Static track uses only 409 repositories (approx. 39k tasks). The observed performance gain of `\codeloraev0{}{}{}` over static baselines could be driven by the sheer scale of the training data rather than the GRU-based sequential modeling. A robust scientific claim requires an ablation study where the Evolution model is trained on a subsampled dataset matching the Static track’s size, or vice versa, to isolate the architectural contribution.

Third, the claim that `\codelorastatic{}{}{}` matches or exceeds the per-repo LoRA (pLoRA) upper bound on In-Repo tasks (66.2% vs 64.0% EM) is statistically fragile. Appendix B notes high variance in pLoRA performance (sigma=20.9%). Without reporting confidence intervals or performing a paired statistical test (e.g., Wilcoxon signed-rank) across the 389 repositories, it is impossible to determine if this “matching” is a genuine breakthrough or a statistical fluctuation. The current evidence does not robustly support the conclusion that the hypernetwork fully captures the per-repo signal without the risk of overfitting.

To strengthen the paper, the authors must address these statistical and control issues before the central claims regarding generalization and upper-bound performance can be accepted.

Action items.

- **[science]** The OOD evaluation (Table 3, Sec 5.3) is confounded by target length artifacts (median 7 chars vs 12-13). While the authors acknowledge this in the Limitations, the main

results section claims a definitive lead (+1.8pp) without statistical significance testing or a length-normalized metric (e.g., EditSim) to isolate the model’s generalization capability from the shorter answer bias.

- **[science]** The Evolution track training set (45,516 commits) is orders of magnitude larger than the Static track (409 repos). The paper does not provide a controlled ablation showing whether `\codeloraevo\{}`’s superior performance stems from the GRU architecture or simply the massive increase in training data volume. A fair comparison requires matching the number of training examples or demonstrating performance saturation.
- **[science]** The claim that `\codelorastatic\{}` matches the per-repo LoRA (pLoRA) upper bound on In-Repo tasks (66.2% vs 64.0% EM) is counter-intuitive and lacks rigorous statistical validation. Given the variance in per-repo performance ($\sigma=20.9\%$ for pLoRA), the authors must report confidence intervals or a paired statistical test (e.g., Wilcoxon signed-rank) to confirm this is not a random fluctuation or overfitting artifact.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical rigor of the evaluation requires strengthening to fully support the claims of superiority, particularly regarding the Out-of-Distribution (OOD) results and the magnitude of performance gains.

First, the paper reports point estimates for performance metrics (e.g., Exact Match percentages) without accompanying measures of uncertainty. In Table~\ref{tab:ood_results}, `\codeloraevo\{}` leads `\codelorastatic\{}` by 1.9 percentage points (74.1% vs 72.2%) and `\texttt{sLoRA\{}` by 1.8 points. Similarly, in the static track (Table~\ref{tab:main_results}), the gain over the best baseline is substantial. However, the manuscript does not report standard deviations, confidence intervals, or results from statistical significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) across multiple random seeds or data splits. Given the high variance observed in per-repo LoRA performance (median 62.5%, $\sigma = 20.9$ as noted in Appendix~\ref{app:per_repo_variance}), it is critical to demonstrate that the observed improvements are statistically significant and not artifacts of random initialization or specific data splits.

Second, the OOD evaluation presents a known confound: the OOD targets are significantly shorter (median 7 chars vs 12–13 chars), which the authors admit inflates Exact Match scores. While the authors note that `\codeloraevo\{}` still leads by a narrower margin, the primary claim of “superior generalization” relies heavily on the EM metric. To robustly support the conclusion that the model generalizes better rather than simply exploiting the shorter target length, the authors should re-evaluate the OOD results using length-agnostic metrics (EditSim, CodeBLEU) as the primary evidence, or explicitly normalize the EM scores. Currently, the reliance on EM

in the presence of a known distribution shift weakens the statistical validity of the OOD claim.

Finally, the experimental protocol for model selection needs clarification. The text states that the “best checkpoint [is selected] by CR-val loss.” It is essential to confirm that the test sets (CR Test, IR Test, OOD Test) were strictly held out and never used for hyperparameter tuning or early stopping decisions. If the test set influenced any part of the training loop or model selection, the reported performance metrics would be optimistically biased. The manuscript should explicitly state the separation of train/validation/test splits to ensure reproducibility and validity.

While the dataset statistics (Table~\ref{tab:dataset_actual}) and error analysis (Appendix~\ref{app:errors}) are descriptive and well-presented, the inferential statistics required to validate the comparative claims are missing.

Action items.

- **[science]** Report statistical significance (e.g., p-values, confidence intervals, or effect sizes) for the reported performance gains (e.g., the +1.8pp OOD lead and +4.8pp CR lead). Without variance estimates or significance testing, it is unclear if these improvements are robust or due to random seed variance.
- **[science]** Clarify the experimental protocol for hyperparameter selection. The paper reports best checkpoint selection by ‘CR-val loss’ but does not specify if the test set was used for any tuning or if a strict hold-out validation set was maintained throughout. Ensure the test set was never used for model selection to prevent data leakage.
- **[science]** The OOD evaluation section explicitly notes that target lengths are shorter in the OOD set, inflating Exact Match (EM) scores. However, the primary claim of superiority relies on EM. Provide a re-analysis of the OOD results using length-normalized metrics (e.g., EditSim or CodeBLEU) as the primary comparison to ensure the conclusion is not an artifact of the target length distribution shift.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a technically sound contribution, but the writing quality is significantly compromised by the presence of unresolved LaTeX macros and structural redundancy.

First, the most critical issue is the extensive use of unresolved macros (e.g., `\UseMacro{cr-em-codelorastatic}`, `\UseMacro{num-repos-total}`) throughout the Introduction, Results, and Conclusion sections. In their current state, these placeholders render the prose unreadable and obscure the specific quantitative claims the authors are making. For instance, in Section 5, sentences like “`\codelorastatic` achieves `\UseMacro{cr-em-codelorastatic}`% CR EM” fail to communicate the actual performance metrics. These must be resolved to their numeric values

(e.g., “63.8%”) to ensure the text is self-contained and verifiable.

Second, the document structure suffers from significant duplication. The main body (Sections 1–7) covers the Introduction, Method, Benchmark, Setup, Results, and Conclusion. However, the Appendix (starting around e002) repeats large portions of the Results, Conclusion, Limitations, and Dataset Details sections. For example, the “OOD Evaluation Caveats” and “Broader Analysis” appear to be re-introduced with similar text and tables already present or implied in the main body. This redundancy disrupts the narrative flow and suggests a compilation artifact where draft sections were not properly pruned. The authors should consolidate these sections, ensuring that the main body contains the primary narrative and the appendix is strictly reserved for supplementary details not essential to the main argument.

Finally, there are minor inconsistencies in figure referencing. In the Appendix (e001), the label `\label{fig:architecture_evo}` is placed after a block of text describing the architecture steps, rather than immediately following the figure environment or caption. While this may not break compilation, it is non-standard and risks broken references in the final PDF.

Addressing these issues—resolving all macros, removing duplicate sections, and standardizing label placement—will significantly improve the readability and professionalism of the paper.

Action items.

- **[writing]** In Section 5 (Results), the text repeatedly uses LaTeX macros (e.g., `\UseMacro{cram-codelorastatic}`) instead of resolved numeric values. This renders the prose unreadable and prevents verification of claims. Replace all such macros with their actual numeric values before final submission.
- **[writing]** The manuscript contains duplicate sections. Section 6 (Results) and Section 7 (Conclusion) appear in the main body, but identical or near-identical content (including Results, Conclusion, Limitations, and Dataset Details) is repeated in the Appendix (Sections starting at e002). Consolidate these into a single, coherent flow to avoid redundancy.
- **[writing]** In the Appendix (e001), the text describes Figure 1 (architecture) but the label `\label{fig:architecture_evo}` is placed after a paragraph of text rather than immediately following the figure environment or caption, which may cause reference errors in the compiled PDF.